

Akif Zahin

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TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS, PHP

Frameworks: React, Next.js, Node.js, FastAPI, TensorFlow, PyTorch, OpenCV

Developer Tools: Git, VS Code, Android Studio, Docker

Libraries: pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, HuggingFace

EXPERIENCE

Guardian Life Insurance Ltd | *Intern, Information Technology Department* May 2026 – Present

- Rebuilt internal phonebook system using **Next.js**, **TypeScript**, and **Tailwind CSS**, improving usability and maintainability
- Revamped company website frontend with **React**, enhancing UI consistency and user experience
- Collaborated with cross-functional teams to architect and implement **MVC** design patterns for upcoming internal tools
- Designed and managed relational database schemas using **MySQL** to support new application features

PROJECTS

akifzahin.dev | *Astro, TypeScript, Turso (libSQL), Tiptap, JWT* May 2026 – Present

- Designed and built a full-stack **CMS from scratch** to power a personal portfolio and blog
- Implemented a **rich-text editor** (Tiptap) with autosave, image uploads, and draft/publish workflows
- Built a comment moderation pipeline with an admin approval queue and threaded replies
- Designed relational schema (Turso/libSQL) for posts, comments, and visitor-scoped engagement (claps, likes) without requiring user accounts

Guardian Phonebook | *Next.js, TypeScript, Keycloak, Tailwind CSS* May 2026 – July 2026

- Built a production employee directory replacing a legacy static HTML phonebook, deployed at **Guardian Life Insurance Ltd**
- Integrated **Keycloak SSO** for authentication, migrating legacy auth from SHA-1 to OAuth 2.0
- Designed an admin dashboard with scheduled **LDAP sync** to keep the public directory current
- Built as part of a company-wide internal tooling initiative during IT internship

NewLens | *Python, FastAPI, PyTorch, HTML, JavaScript, Docker* July 2024 – Dec 2024

- Engineered a full-stack web app for real-time sentiment and political bias detection in news articles
- Fine-tuned a **BERT-based transformer model** via HuggingFace to classify bias as **Left, Right, or Neutral**
- Implemented **FastAPI REST endpoints** for model serving with asynchronous I/O for low latency
- Containerized the entire system using **Docker** for reproducible deployment
- Part of junior design project (**CSE299**)

LM-FCIL | *Python, Kotlin, TensorFlow, TFLite, Federated Learning, OpenCV* Jan 2025 – Aug 2025

- Developed an Android app for **offline handwriting segmentation and recognition** using deep CNNs and OpenCV
- Implemented a **Federated Learning** framework for on-device model updates without sharing raw data
- Used **MobileNetV3 Small** as backbone; designed client-server sync protocol for decentralized training
- Part of senior research project (**CSE498R**)

TimeGaitNet | *Python, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas* June 2025 – Dec 2025

- Designed a deep learning pipeline for **Parkinson's Freezing of Gait (FoG) detection** using wearable sensor data
- Leveraged **multi-task learning** and multimodal fusion of temporal and spatial gait features
- Applied extensive data augmentation to mitigate class imbalance; first novel use of state space models for FoG
- Part of senior capstone project (**CSE499B**)

EDUCATION

North South University

Bachelor of Science in Computer Science and Engineering — CGPA: 3.45

Dhaka, Bangladesh

Oct 2021 – Dec 2025

South Breeze School Uttara

O-Levels & A-Levels

Dhaka, Bangladesh

June 2019 – June 2021

- **O-Level Grades:** four 8s and four 9s
- **A-Level Grades:** two A*s and two As

AWARDS

Full Merit Based Entrance Scholarship — North South University, 2021

Daily Star Award — O-Level Academic Excellence, 2019